



Somashekhar Dehury

Roll No.: B210610EC

BTech

Electronics and Communication Engineering

in Somashekhar Dehury

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National Institute Of Technology, Calicut

EDUCATION

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| •National Institute of Technology, Calicut | <i>Dec 2021 - May 2025</i> |
| <i>B.Tech in Electronics & Communication Engineering</i> | CGPA: 8.95 |
| •Viswaprakash Central School | <i>2020</i> |
| <i>Intermediate(CBSE)</i> | Percentage: 96.8% |
| •VSSC Central School | <i>2018</i> |
| <i>Matriculation(CBSE)</i> | Percentage: 97.8% |

EXPERIENCE

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|---|----------------------------|
| •DAAD WISE Scholarship - GSI Helmholtz centre | <i>May 2024 - Jul 2024</i> |
| <i>Intern / Dipl.-Ing. Joachim Maurer</i> | Darsmtadt-Germany |
| – Research intern at GSI Helmholtz centre for heavy ion Research | |
| – Programmed different microcontroller platforms like Arduino, ESP32 with Arduino IDE as well as storing data to InfluxDB with microcontroller. | |
| – Acquired and visualized data on displays by connecting to power supplies via serial interface and controlling them. | |
| •Indian Space Research Organisation | <i>May 2023 - Jun 2023</i> |
| <i>Intern / Dr Gopal Jee</i> | Trivandrum |
| – Simulated the control system of the Lunar lander using Matlab | |
| – Explored using Deep Learning Techniques for automatic crater detection for the lander | |

PERSONAL PROJECTS

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|---|----------------------------|
| •Command Controlled car | <i>Mar 2023 - Apr 2023</i> |
| <i>Microcontroller and Microprocessor project</i> | |
| – Designed the control system of the car using 8051 microcontroller, coded in C and connected to mobile phone using the wifi module. | |
| – Controlled the motion of car in the form of small alphabetical commands. | |
| – TechStacks: C, 8051 microcontroller, Keil, wifi module(esp8266) | |
| •Mobile Phone Detector | <i>Oct 2022 - Nov 2022</i> |
| <i>Networks Lab(Tested the circuit of the detector for debugging)</i> | |
| – Designed the circuit using proteus. | |
| – Simulated the circuit on LtSpice and successfully implemented on the breadboard after testing. | |
| – TechStacks: Proteus, LtSpice, Opamp(CA3130), Transistor(BC548) | |
| •Automatic Pothole Filler | <i>Apr 2023</i> |
| <i>Ramaiah Hackathon(Worked in the coding & integrating part of the Arduino & Wifi module)</i> | |
| – Developed the prototype of a pothole filling machine using ML model. | |
| – Implemented the detection of potholes in realtime and filling of the pothole. | |
| – Designed the control system of the machine using Arduino Uno and Wifi module(esp8266)(We were selected as finalists for the finale round) | |
| – TechStacks: Arduino, Machine Learning, YoLo(V8), Wifi Module, Python | |
| •Automatic Book Scanner | <i>Jan 2022 - Aug 2022</i> |
| <i>Smart India Hackathon 2022(Worked in both the hardware and AI, ML part of the project)</i> | |
| – Designed and developed an automatic book scanning machine. | |
| – Executed the automatic control of turning of pages of a book and converting the hardcopy of the text to softcopy. | |
| – Implemented pretrained NLP models for translation of texts to different languages.(We were the only team from the Kerala region called by the SIH as finalists,held in New Delhi) | |
| – TechStacks: Deep Learning(NLP), Arduino | |

TECHNICAL SKILLS AND INTERESTS

Languages: C, C++, Python

Developer Tools: VS Code, Matlab, LtSpice, SciLab, InfluxDB, Jupyter Notebook

Frameworks: GitHub, Keil, Arduino, OpenCV, TensorFlow(Basics)

Soft Skills: Communication, Problem Solving, TeamWork, Work Ethic, Project Management

Coursework: Microprocessors and Microcontrollers,Basics of Computer Programming, Probability & Statistics

Areas of Interest: Robotics, Arduino, Embedded Systems

CONFERENCES & WORKSHOPS

•**IISC Bangalore** Artificial Intelligence in Precision Medicine *Jan, 2024*

•**IIT Jodhpur** Primer on 3D Vision in AI *Dec, 2023*

CERTIFICATIONS

•**DeepLearnig.AI** Supervised Machine Learning: Regression & Classification *Oct, 2022*

•**NIT Warangal** Python Programming for Artificial Intelligence and Machine Learning *Nov, 2021*

ACHIEVEMENTS

•**DAAD WISE Scholar** Research intern at GSI Helmholtz centre Darmstadt, Germany *2024*

•**Qualified PRMO** Qualified PRMO olympiad twice for the next round RMO *2018, 2020*

•**Gold Medallist** Mathematics Olympiad(Sof) *2018*

•**Silver Medallist** Science Olympiad(Sof) *2018*